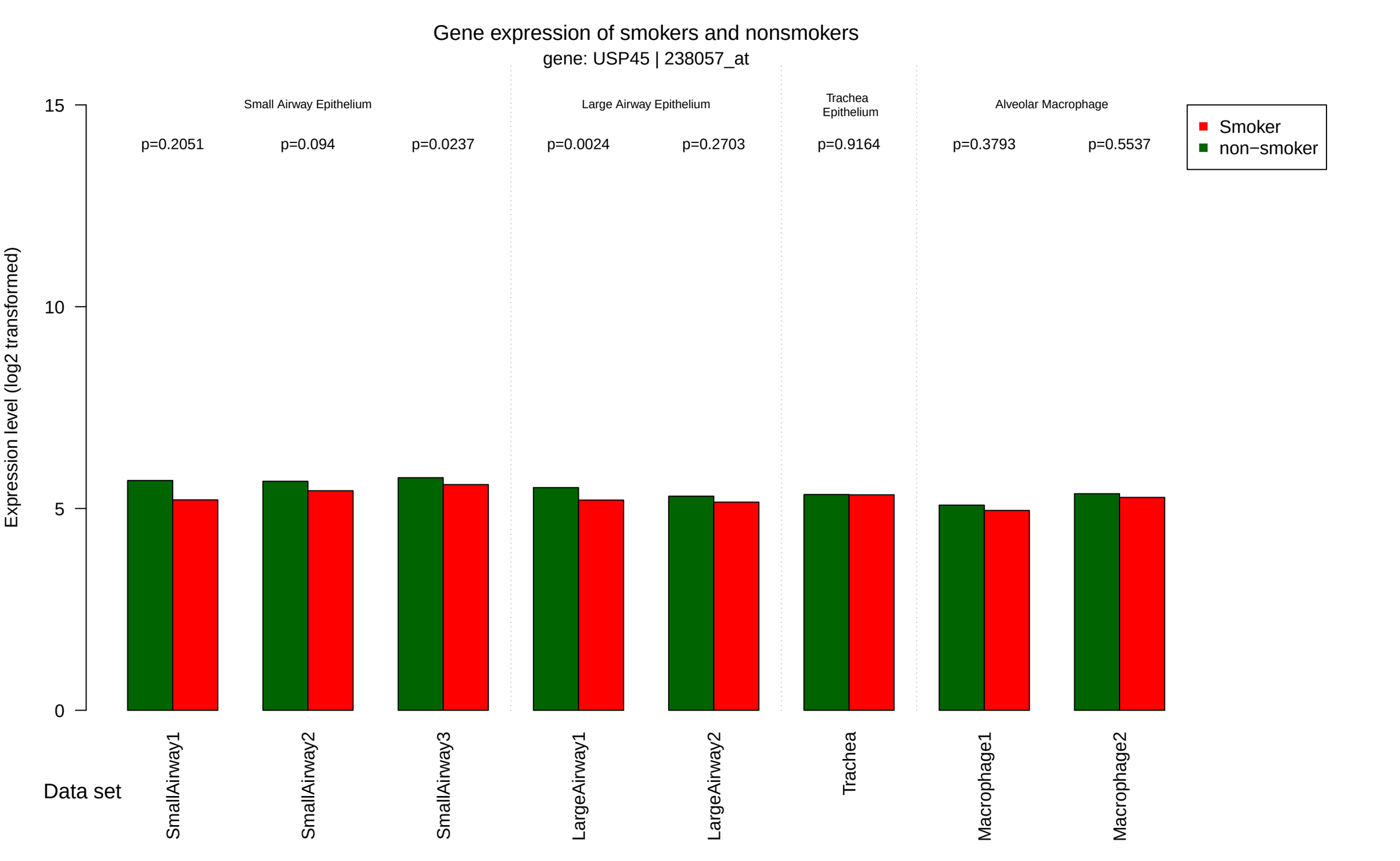
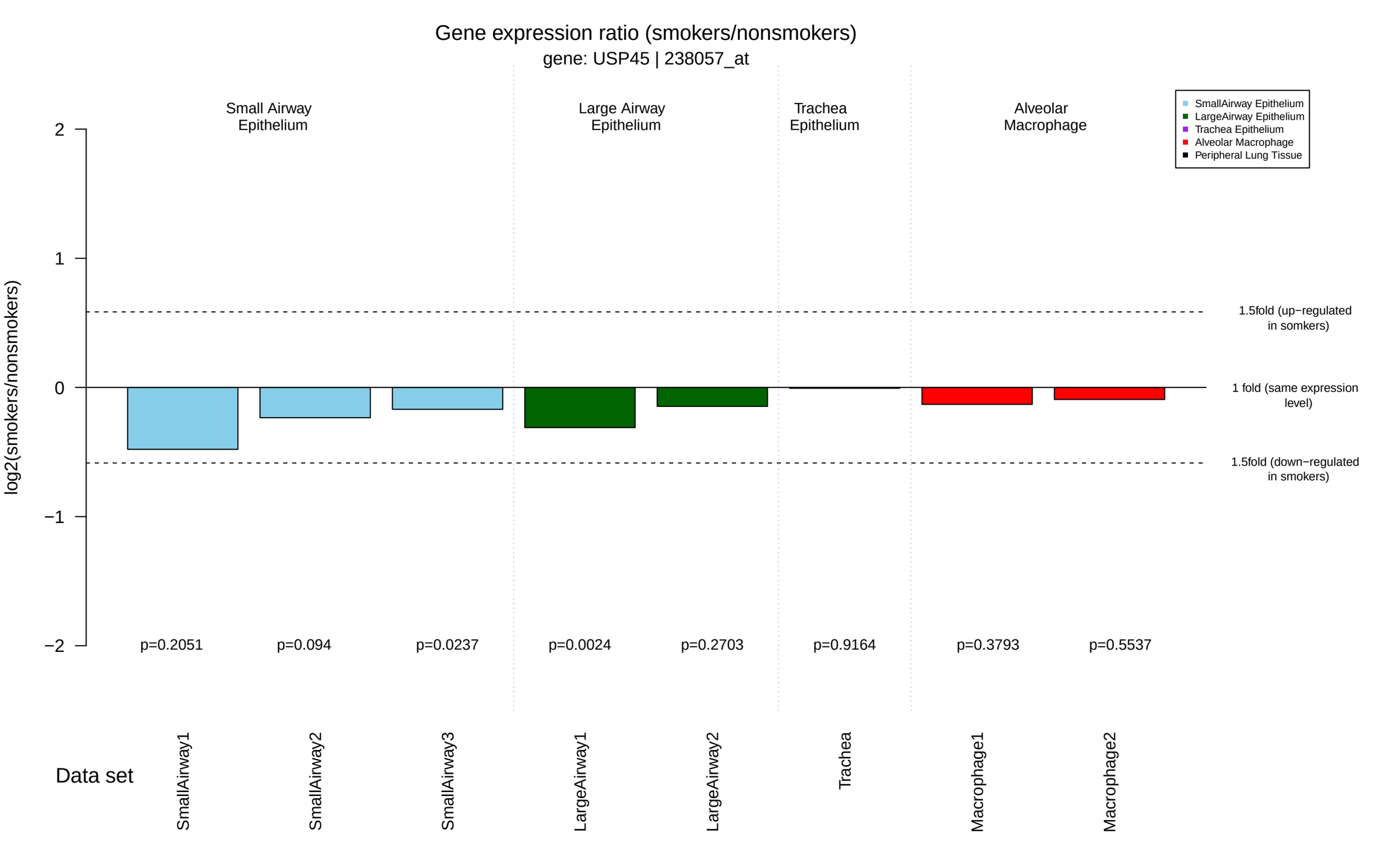
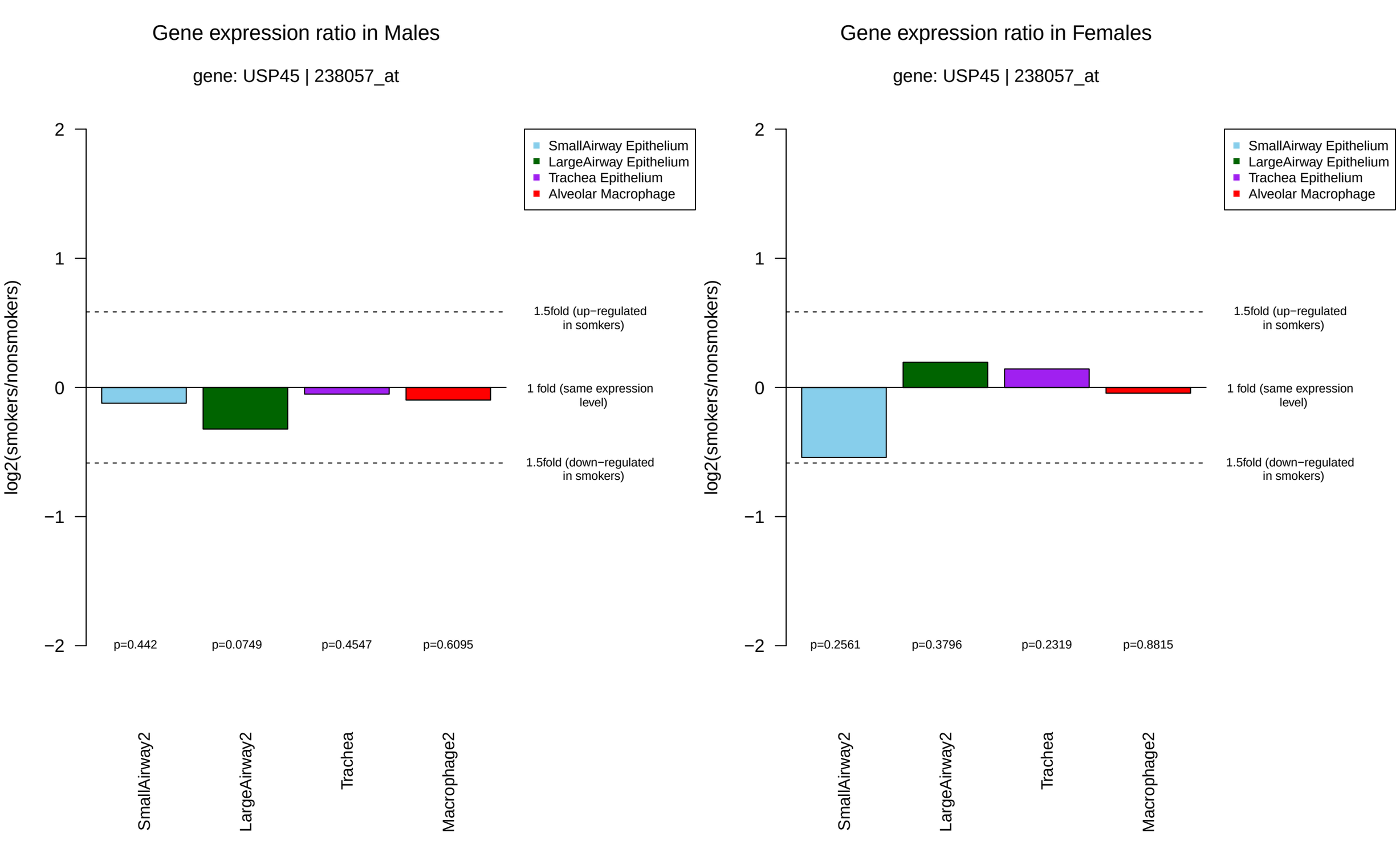
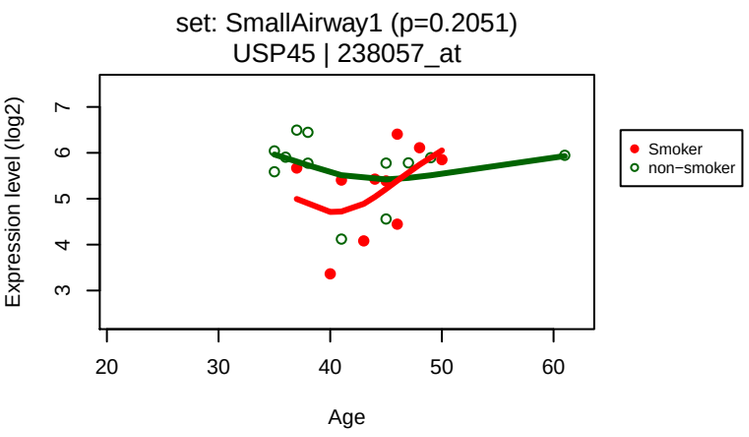
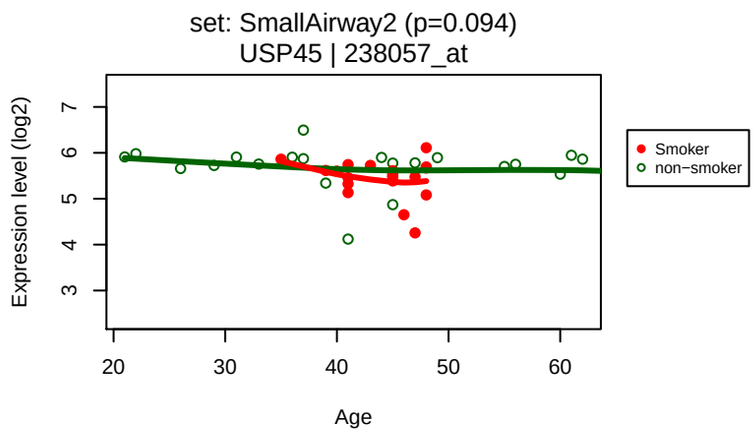
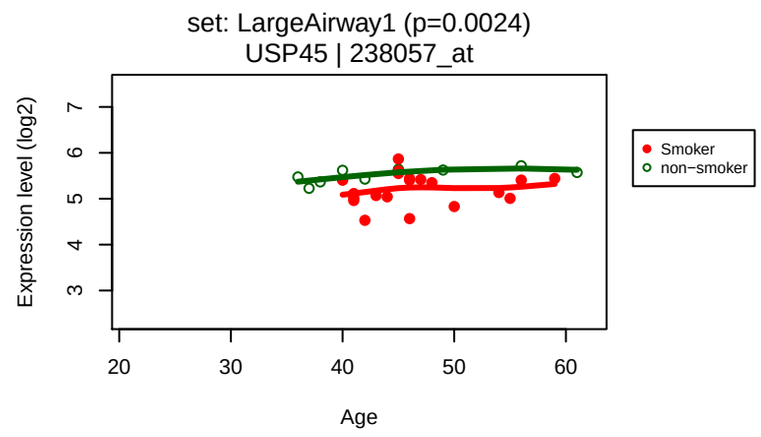

Gene expression of smokers and nonsmokers
gene: USP45 | 238057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2051
p=0.094
p=0.0237
p=0.0024
p=0.2703
p=0.9164
p=0.3793
p=0.5537
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

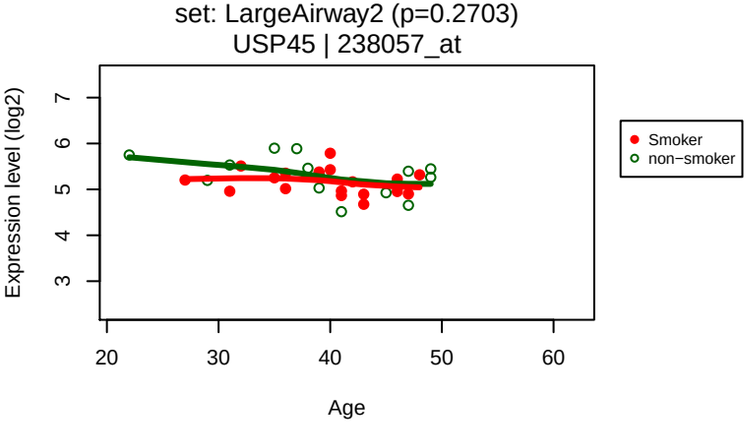

Gene expression ratio (smokers/nonsmokers)
gene: USP45 | 238057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2051
p=0.094
p=0.0237
p=0.0024
p=0.2703
p=0.9164
p=0.3793
p=0.5537
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

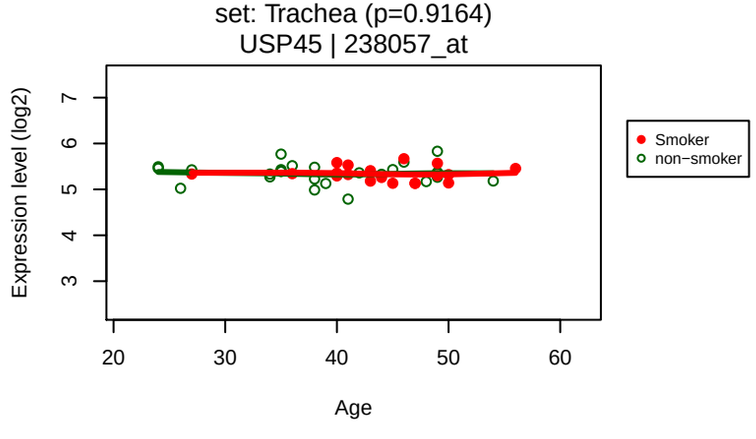

Gene expression ratio in Males
gene: USP45 | 238057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.442
p=0.0749
p=0.4547
p=0.6095
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP45 | 238057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2561
p=0.3796
p=0.2319
p=0.8815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

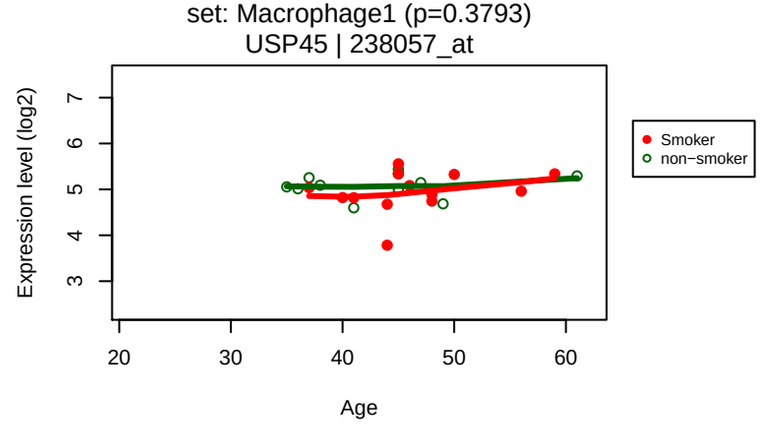

set: SmallAirway1 (p=0.2051)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker

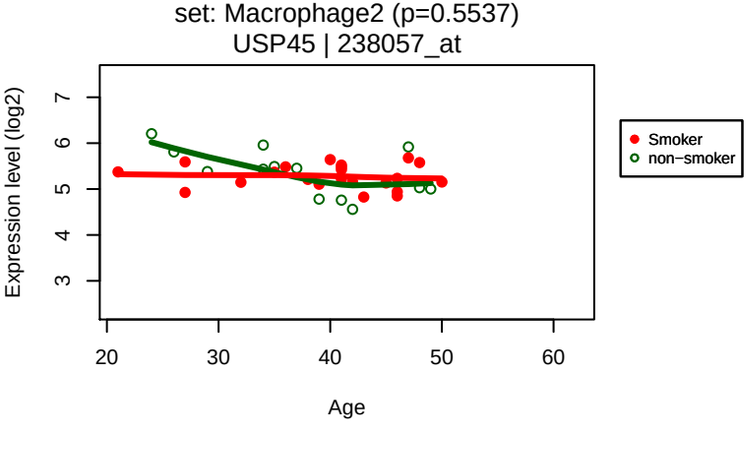

set: SmallAirway2 (p=0.094)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker

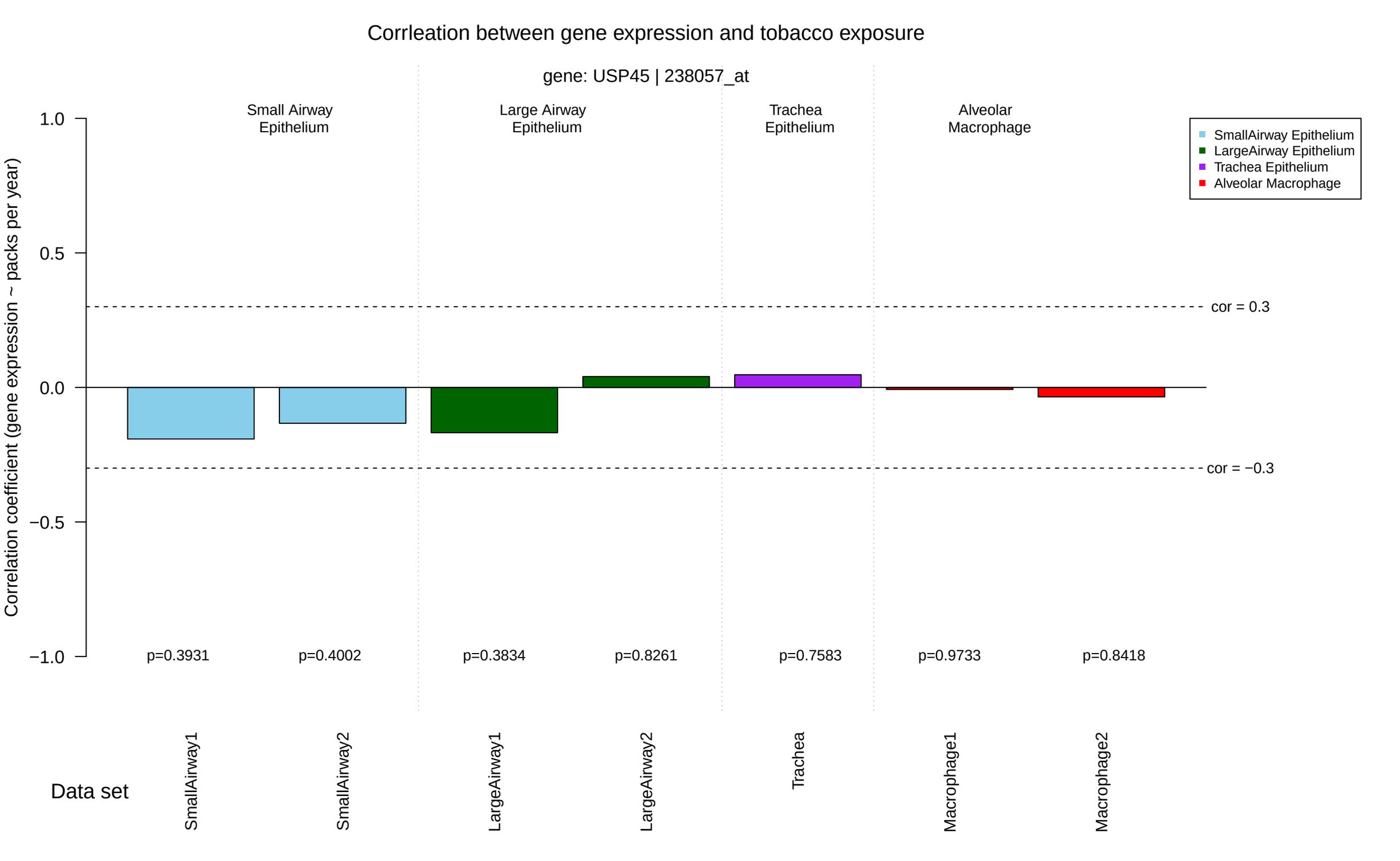

set: LargeAirway1 (p=0.0024)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2703)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9164)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3793)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5537)
USP45 | 238057_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP45 | 238057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3931
p=0.4002
p=0.3834
p=0.8261
p=0.7583
p=0.9733
p=0.8418
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2